

CIT2C18 - Mobile App Development Project Specification AY2026/2027 April Semester

Introduction

This is an **individual** project. The purpose of this project is to understand and apply the knowledge of mobile application development. The project consists of three parts and involves developing a working mobile application using the Flutter framework.

1. Part 1 involves developing a proposal for your mobile application, including a high-fidelity wireframe.
2. Part 2 involves developing a Flutter application based on the proposed design, including UI design, navigation, forms and user authentication-related functionalities.
3. Part 3 involves enhancing the application by implementing database functionalities, one applied AI feature, personalisation and additional features.

Students must implement features using constructs covered in class (e.g. Riverpod, Firebase, Form-related widgets, Navigation-related widgets, etc). The use of external resources, including AI-generated content, must be declared, understood and explained clearly during assessment.

AI tools may be used for debugging, consultation and learning support, but must not be used to generate entire or substantial parts of the application. Students who are unable to explain their code or demonstrate clear understanding of their implementation may receive significant mark deductions, regardless of whether the application is functional.

Part 1 – Proposal (20%)

You are required to submit a proposal for the mobile application that you will develop in Part 2 and 3. There is no restriction on the nature of the application, but it must fulfil the project requirements stated in Part 2 and 3.

As part of your UI design process, you'll use DALL·E (<https://openart.ai/create>), an AI-powered image generation tool, to generate visual concepts such as screen layouts and colour schemes. These ideas can help you explore different design options and guide your final design decisions.

The proposal (refer to LMS for template) must include the following minimum requirements:

Application Description	Provide a name for your application e.g. Sustainability Buddy. Identify your target audience. Provide a list of proposed functions with brief descriptions, including authentication features, ONE* set of CRUD and ONE applied AI feature. Identify one suitable scenario where NETS QR payment can be integrated into your application. * Please refer to the marking rubrics for Part 3. Adding extra CRUD operations will NOT earn additional marks, so keep to just ONE set.
Database Design	Data dictionary* and Entity-Relationship diagram. * Minimum of 2 tables (Users table + one other table). The tables do NOT need to be normalised.
Design Analysis and Wireframe	Provide an analysis of three DALL·E-generated visual concepts, explaining the design elements you selected or refined and the reasons for your choices. High-fidelity design screenshots the application* * Minimum of 5 screens. Authentication related screens (registration, login, forgot password) are considered as ONE screen. Modal forms can be considered separate screens.
Widget Tree	Widget Tree for the home screen* of the application. * Home screen refers to the screen that is NOT authentication related where user accesses and interacts with the main features of the application. It is usually the screen that is shown when the user logs in successfully or when the application launches.
Additional Features	Describe at least one additional functional enhancements* you might implement - these functions should NOT be CRUD related. * Please refer to the Additional Features table in Part 3 of the Project Specifications for some ideas on possible functionalities.

You are free to be as creative as you can but do ensure that the mobile application can be completed within the specified timeframe and it does NOT include gambling, prohibited content or anything inappropriate. You are encouraged to discuss your mobile application ideas with your tutor and submit a project proposal report by **Week 5 (22 May, Friday, 5 PM)**.

Do NOT propose any form of Expense Tracker application, as that is the learning application used in class. Some suggested applications include but are not limited to the following:

Sample Applications	Features
Habit & Wellness Tracker	• CRUD domain: Daily habits (e.g. sleep, water intake, exercise, reading) • Possible applied AI feature: AI-generated habit suggestions and streak analysis • Possible additional feature: Local notifications
Study Planner	• CRUD domain: Study tasks / revision schedules • Possible additional feature: AI-generated study timetable or revision plan based on exam dates and weak subjects • Possible additional feature: Calendar Integration
Food Expiry Tracker	• CRUD domain: Food items • Possible applied AI feature: AI analyses food expiry patterns and predicts which items are likely to expire soon and suggests priority consumption order. • Possible additional feature: Local notifications
E-Commerce Shopping App	• CRUD domain: Orders • Possible applied AI feature: Suggests products based on browsing and purchase history. • Possible additional feature: Push notifications
Travel Planner App	• CRUD domain: Trips • Possible applied AI feature: Suggests travel plans based on budget, duration, etc • Possible additional feature: Flutter Map

Submission Instructions

Submit your Microsoft Word document into TP LMS under the Assessments folder. Your files should be named according to the following format:

Part1_YourName_StudentID_YourClass.docx
e.g.: Part1_JohnTan_1234567D_P01.docx

Please ensure that you have a backup copy of your deliverables in case there is a problem with the online submission.

Part 2: Basic Application and Authentication – Program Code (30%) and Presentation (10%)

Your task is to develop a Flutter application that aligns with the design proposed in Part 1. The application must implement UI design, navigation, forms and authentication features using concepts covered in class.

Your application must apply the knowledge that you have learnt and include the following:

- All proposed screens with navigation using concepts covered in class
 - CRUD-related screens do not need to be functional at this stage
 - E.g. Navigation drawer, Bottom navigation, Tabbed layout
- Use widgets the following widgets
 - Form components: Form, TextFormField and ElevatedButton
 - Layout components: Row and Column
 - Data display: ListView or GridView
- UI Components and Event Handling
 - Form validation and feedback for authentication-related screens
- User Authentication (using Firebase Authentication)
 - To implement basic authentication (register, login, logout, forget password)
 - To explore account management features not covered in class (E.g. change password, email verification)
 - To explore authentication methods not covered in class (E.g. Google Sign-in)
- Implementation of Riverpod for state management
- Good programming practices
 - Well-structured project with appropriate folders and good naming convention as covered in class and well commented code.

* The implementation of complex widgets, such as a calendar that requires external plugins, is not mandatory and may be replaced with an image for Part 2.

You are required to prepare a Word document that includes high-fidelity design screenshots from Part 1, along with screenshots of your implemented Flutter application. Besides submitting the Word document and project source codes on **Week 12 (6 Jul, Monday, 9AM)**, you are to demonstrate the navigation flow during your assignment evaluation, which will be held on **Week 12 (Week of 6 Jul) - Timings to be advised by your tutor.**

Submission Instructions

Submit the entire project folder as a zipped file and the Word document into TP LMS under the Assessments folder. Your zipped document file should be named according to the following format:

Part2_YourName_StudentID_YourClass.zip
e.g.: Part2_JohnTan_1234567D_P01.zip

Part2_YourName_StudentID_YourClass.docx
e.g.: Part2_JohnTan_1234567D_P01.docx

Please ensure that you have a backup copy of your deliverables in case there is a problem with the online submission.

Part 3: Database Operations and Additional Features – Program Code (30%) and Presentation (10%)

You are required to enhance the Flutter application developed in Part 2 by adding data persistency, one applied AI feature, personalisation and additional features.

Your application must apply the knowledge that you have learnt and include the following:

- Part 2's implementation must be fully included and functional
- UI Components and Event Handling
 - Form validation and feedback are present on all screens
- Data Persistency and CRUD operations (using Google Cloud Firestore)
 - To implement ONE set of basic CRUD
 - To explore advanced queries (E.g. Filter, sort, aggregation, etc)
- Applied AI feature that enhances the functionality of the application
 - E.g. Image scanning, chatbot, text-to speech, speech-to-text, etc
- App Personalisation (App Icon, Splash Screen, Themes)
- Implementation of Riverpod for state management
- Good programming practices
 - Well-structured project with appropriate folders and good naming convention as covered in class and well commented code.

The following topics that you have learnt can be implemented as simple additional features:

- Web Service consumption
- Enhanced NETS Payment implementation
- Camera/ Image Gallery access

Some suggested additional features include but are not limited to the following:

Sample Additional Features – Free to use	
Calendar	https://pub.dev/packages/table_calendar
Social Sharing/ Launch weblink in-app	https://pub.dev/packages/flutter_share_me https://pub.dev/packages/url_launcher
Local Notifications	https://pub.dev/packages/flutter_local_notifications
QR Scanner	https://pub.dev/packages/qr_code_scanner
Biometric Login	https://pub.dev/packages/local_auth
Flutter Map	https://pub.dev/packages/flutter_map
Offline Mode Detection and Handling	https://pub.dev/packages/connectivity_plus

Note that the use of paid services or any features requiring credit card information (including but not limited to Google Storage and Google Maps) are not allowed as additional features.

Besides submitting project source codes on **Week 17 (10 Aug, Monday, 9AM)**, you are to present and demonstrate your work during your assignment evaluation, which will be held on **Week 17 (Week of 10 Aug) - Timings to be advised by your tutor.**

Prepare a PowerPoint presentation that includes, but is not limited to, the following:

- Name, matriculation number and class
- Implemented functionalities (including Additional features)
- Video Recording of Application (YouTube link – max 5 mins)

Submission Instructions

Submit the entire project folder as a zipped file and the PowerPoint slides into TP LMS under the Assessments folder. Your zipped document file and PowerPoint slides should be named according to the following format:

Part3_YourName_StudentID_YourClass.zip

e.g.: Part3_JohnTan_1234567D_P01.zip

Part3_YourName_StudentID_YourClass.pptx

e.g.: Part3_JohnTan_1234567D_P01.pptx

Please ensure that you have a backup copy of your application in case there is a problem with the online submission. Please submit the above by **Week 17 (10 Aug, Monday, 9AM).**

File Submission and Backup Policy

When submitting assignments, each student is responsible for ensuring that they do not submit an empty zip file or a corrupted file.

Students must also ensure they are using a suitable device when working on assignments / projects, maintain proper backups of their work and do not cite 'laptop issues' as a reason for assignment extensions as such reasons will not be considered.

Penalty for Late Submission

- **Late within 1 day:** 10% deduction from absolute mark given for the assignment e.g 75 marks (100 marks max) becomes 65 marks (deduct 10% of 100 marks)
- **Late > 1 day and up to 2 days:** 20% deduction from absolute mark
- **Late > 2 days:** 0 marks awarded

Project - Grading Criteria

The grading criteria for Part 1 (20%) will be based on the following:

Criteria	In Context	Performance Level				
		Excellent	Good	Average	Poor	Below Standard
Project Proposal (20%)	Application Description (5%)	Proposal write-up includes ALL of the following: - application name, - target audience identified clearly, - full set of authentication features, - full set of CRUD, - applied AI feature, - possible scenario for NETS QR payment Comprehensive, clear and meaningful description of proposed features described with aid of diagrams or some form of visual aids.	Proposal write-up includes ALL of the following: - application name, - target audience identified clearly, - full set of authentication features, - full set of CRUD, - applied AI feature, - possible scenario for NETS QR payment Comprehensive and clear description of the proposed features.	Proposal write-up includes MOST of the following: - application name, - target audience identified clearly, - full set of authentication features, - full set of CRUD, - applied AI feature, - possible scenario for NETS QR payment Proposed features described but lacks clarity or depth.	Proposal write-up includes SOME of the following: - application name, - target audience identified clearly, - full set of authentication features, - full set of CRUD, - applied AI feature, - possible scenario for NETS QR payment Description of proposed features is vague, poorly described, or not present.	Proposal write-up is incomplete, non-submission Description of proposed features is incomplete, or non-submission, or Clear evidence of plagiarism detected (disciplinable offence)
	Database Schema (5%)	Extensive ER Diagram that contains appropriate tables and constraints that fully supports the proposed functions. All data items in the tables are elaborated with more details with the aid of their respective data dictionary.	ER Diagram that contains mostly correct tables and constraints that supports most of the proposed functions. All data items in the tables are elaborated with more details with the aid of their respective data dictionary.	Partial ER Diagram that contains some correct tables and constraints used that supports most of the proposed functions. Some data items in the tables are elaborated with some details with the aid of their respective data dictionary.	Sparse ER Diagram and/or data dictionary that contains few correct tables and constraints that supports the proposed functions and/or lack in details.	Database Schema are incomplete, or Non-submission of database schema or clear evidence of plagiarism detected (disciplinable offence)

	Reflection on Design choice (1.5%)	Provides a thorough and insightful reflection on three DALL-E-generated visual concepts and design choices, linking them clearly to user experience, usability, aesthetics, and app goals.	Provides a clear reflection with good reasoning on three DALL-E-generated visual concepts and design choices, considering most aspects of usability and aesthetics.	Reflection is basic but addresses key aspects like usability and aesthetics; reasoning lacks depth.	Superficial reflection with minimal reasoning on the design choices made.	No meaningful reflection; does not explain or justify design choices, or Non submission or clear evidence of plagiarism detected (disciplinable offence)
	Design Screenshots (hi-fi) (5%)	The screenshots clearly and fully illustrate functions as highlighted in the project scope. Contains at least 5 screens. Design is Intuitive, easy to learn, and easy to use.	The screenshots clearly and fully illustrate functions as highlighted in the project scope. Contains at least 5 screens. Design is easy to use once you learn it.	The screenshots partially illustrate functions as highlighted in the project scope. Contains at least 5 screens. Design is easy to use once you learn it.	The screenshots barely illustrate functions as highlighted in the project scope. Contains 4 screens or less. Design is easy to use once you learn it.	The screenshots are incomplete, or Non-submission of hi-fi design screenshots or clear evidence of plagiarism detected (disciplinable offence)
	Widgets Tree (1%)	Widget tree of main screen is complete and mostly correct.		Widget tree of main screen is incomplete or mostly incorrect.		No widget tree proposed, or Non submission or clear evidence of plagiarism detected (disciplinable offence)
	Additional Feature (2.5%)	Feature is clearly described, relevant, and non-CRUD Technically feasible within the project scope Clear and well-explained integration with the application	Feature is clearly described and non-CRUD Feasible within the project scope Adequate explanation of how it integrates with the application	Feature is basic or partially unclear, but non-CRUD Feasibility may be questionable Limited explanation of integration	Feature is unclear, trivial, or overlaps with CRUD Poor or vague explanation, with little consideration of integration	No additional feature proposed, or Non submission or clear evidence of plagiarism detected (disciplinable offence)

The grading criteria for Part 2 (40%) – Program Code (30%) and Presentation (10%) will be based on the following:

Criteria	In Context	Performance Level				
		Excellent	Good	Average	Poor	Below Standard
Program Code (30%)	Implemented screens (5%)	5 proposed screens are implemented. Max 2 marks for each screen, breakdown: 1 mark for aligning implemented screen with proposed design in Part 1 1 mark for effective use of white space, graphic elements and/or alignment to organise content.				No screens implemented Clear evidence of plagiarism detected (disciplinable offence)
	Navigation (2.5%)	Navigation is fully implemented using appropriate structures taught in class (e.g. Drawer, BottomNavigationBar, DefaultTabController). Navigation flow is logical, smooth, and bug-free.	Navigation is mostly complete using appropriate structures taught in class (e.g. Drawer, BottomNavigationBar, DefaultTabController). Navigation is functional with minor issues in flow or structure.	Basic navigation is present but limited (e.g. button navigation) or inconsistent.	Navigation is incomplete, confusing, or contains multiple errors.	No navigation implemented or plagiarism detected.
	Widgets (2.5%)	Appropriate use of all of the following widgets taught in class: • Form, TextFormField, ElevatedButton • Row, Column • ListView/GridView All widgets are used consistently throughout the application.	Appropriate use of all of the following widgets taught in class: • Form, TextFormField, ElevatedButton • Row, Column • ListView/GridView Most required widgets are used consistently throughout the application.	Appropriate use of some of the following widgets taught in class: • Form, TextFormField, ElevatedButton • Row, Column • ListView/GridView Some required widgets are used inconsistently throughout the application.	Appropriate use of few of the following widgets taught in class: • Form, TextFormField, ElevatedButton • Row, Column • ListView/GridView Most required widgets are used inconsistently throughout the application.	No meaningful use of required widgets, or Clear evidence of plagiarism detected (disciplinable offence)
	Form validation (2.5%)	Input validations are present in all input fields. Appropriate keyboard types are implemented for all input fields.	Input validations are present in most input fields. Appropriate keyboard types are implemented for most input fields.	Input validations are present in some input fields. Appropriate keyboard types are implemented for some input fields.	Input validations or appropriate keyboard types are present in a few input fields.	Form validation not implemented in mobile application, or Clear evidence of plagiarism detected (disciplinable offence)

	Feedback (2.5%)	Clear and consistent feedback provided (e.g. SnackBars, error messages, success states).	Feedback is implemented and mostly clear, with minor inconsistencies.	Some feedback present but limited or inconsistent.	Few feedback is implemented.	No feedback implemented, or Clear evidence of plagiarism detected (disciplinable offence)
	Program Code (2.5%)	Code is well-structured according to class conventions (e.g. screens, providers, services folders). Consistent and meaningful naming conventions used throughout. Code is clearly commented where necessary. Riverpod is used correctly and effectively, with clear separation between UI and logic (e.g. providers, services).	Code is mostly well-structured with minor inconsistencies. Naming conventions are generally clear. Some useful comments are present. Riverpod is used correctly, but structure or separation of logic could be improved.	Basic project structure is present but inconsistent or incomplete. Naming conventions are sometimes unclear. Limited or inconsistent comments. Basic use of Riverpod, but with weak separation of logic.	Poor or unclear project structure. Inconsistent or inappropriate naming conventions. Little to no meaningful comments. Minimal or incorrect use of Riverpod.	Little or no evidence of good programming practices. Clear evidence of plagiarism detected (disciplinable offence)
	User Authentication (12.5%)	Basic Firebase authentication (register, login, logout and forgot password) implemented with customised settings with no bugs and errors. At least two other account management features and two other authentication methods (which must not be taught in class) is implemented with no bugs and errors.	Basic Firebase authentication (register, login, logout and forgot password) implemented with customised settings with no bugs and errors. At least one other account management features and one other authentication methods (which must not be taught in class) is implemented with no bugs and errors.	Basic Firebase authentication (register, login, logout and forgot password) implemented with no bugs and errors.	Some form of user authentication present but with errors, or At least one simple authentication method implemented with no bugs and errors.	No form of user authentication present, or Clear evidence of plagiarism detected (disciplinable offence)

Presentation (10%)	Live Demonstration (5%)	Demonstrates the application smoothly and completely, covering all key features (e.g. navigation, authentication, main screens). Flow is logical, well-paced, and clearly organised with no major issues.	Demonstration is mostly smooth and complete with minor issues. Most key features are shown with a clear and logical flow.	Demonstration is complete but lacks clarity in flow. Most key features are shown, but others are missing or unclear.	Demonstration is incomplete or disorganised. Only a few features are shown, or there are multiple technical issues.	Student was absent for project presentation and demonstration without a valid reason Clear evidence of plagiarism detected (disciplinable offence)
	Q&A (5%)	Answers all technical/code-related questions confidently, including reasoning behind implementation choices; explains key parts of the code accurately and in detail Present within the given time-limit.	Answers most technical/code-related questions confidently; gives reasonable explanation of code and logic. Present within the given time-limit.	Struggles with some code-related questions; explanations lack depth or clarity.	Unable to respond to most code or technical questions; shows limited understanding of own code.	

The grading criteria for Part 3 (40%) – Program Code (30%) and Presentation (10%) will be based on the following:

Criteria	In Context	Performance Level				
		Excellent	Good	Average	Poor	Below Standard
Program Code (30%)	CRUD operations (5%)	CRUD with Cloud Firestore implemented: Select all, Select one, Insert, Update, Delete (2 marks for each category of database operation that was implemented meaningfully, as proposed, with no errors)				No database operation is present, or Clear evidence of plagiarism detected (disciplinable offence)
	Advanced queries (2.5%)	CRUD with Cloud Firestore implemented: Select with filter criteria (other than identifier), Select with multiple filter criteria (same field), Select with multiple filter criteria (different fields), Select with sort order, Select with aggregation (1 mark for each category of database operation that was implemented meaningfully, max 5 marks)				No advanced queries present, or Clear evidence of plagiarism detected (disciplinable offence)

	Data (1.5%)	Firestore documents contain comprehensive set of meaningful records/items.	Firestore documents contain good set of meaningful records/items.		Firestore documents contain very few records/items.	Non-submission or clear evidence of plagiarism detected (disciplinable offence)
	Form validation and Feedback (2.5%)	Input validations are present in all input fields. Multiple feedback is provided via AlertDialog and SnackBar. Other form of feedback are explored and implemented.	Input validations are present in most input fields. Multiple feedback is provided via AlertDialog or SnackBar	Input validations are present in some input fields. Some feedback is provided via AlertDialog or SnackBar.	Input validations are present in a few input fields. Few feedback are provided via AlertDialog or SnackBar.	Form validation not implemented in mobile application, or Feedback is not implemented in mobile application, or Clear evidence of plagiarism detected (disciplinable offence)
	Applied AI feature (5%)	AI feature is highly relevant, meaningful, and enhances core app functionality. Well-integrated into user flow. Accurate results with smooth user interaction and no errors.	AI feature is relevant and functional. Integration is good but may have minor usability or flow issues. Accurate results with smooth user interaction and no errors.	AI feature works but is basic or loosely integrated into the app. Limited impact on overall functionality. Some errors present when interacting with AI feature.	AI feature is partially implemented, inconsistent, or buggy. Weak integration into the app.	Incomplete or no attempts of any applied AI feature, or Clear evidence of plagiarism detected (disciplinable offence)
	Additional features (7.5%)	Multiple advanced additional features implemented with no bugs and errors. Mobile app is highly responsive. Effective use of white spaces.	Some advanced additional features implemented with no bugs and errors.	At least one advanced additional feature implemented with no bugs and errors.	At least one simple additional feature implemented with no bugs and errors.	Incomplete or no attempts of any advanced feature, or Clear evidence of plagiarism detected (disciplinable offence)
	Competition and commercial viability (2.5%)	Mobile application is complete with comprehensive set of functionalities that has good competition and commercial viability with high relevance to the theme.	Mobile application is complete with comprehensive set of functionalities that has good competition and commercial viability with high relevance to the theme.	Mobile application has good competition and commercial viability with some relevance to the theme but some touching up is required.	Little competition, commercial viability or relevance to theme.	No competition, commercial viability or relevance to theme.

	Personalisation (2.5%)	Mobile application is personalised with relevant and attractive app icon and splash screen. Personalisation via highly customised themes is implemented in the mobile app.	Mobile application is personalised with relevant app icon and splash screen. Personalisation via themes is implemented in the mobile app.	Mobile application is personalised with relevant app icon and splash screen, or Personalisation via themes is implemented in the mobile app.	Some form of personalisation present	Some form of personalisation present but with errors, or Clear evidence of plagiarism detected (disciplinable offence)" Program failed to compile or run successfully.
	App functionality and code quality (1%)	Mobile application is able to run on web browser and emulator with no errors. Code is well commented.	Mobile application is able to run on web browser and emulator with no errors. Code is commented.	Mobile application is able to run on web browser with very few (<3) errors.	Mobile application is able to run on web browser with some (3 to 5) errors.	Program failed to compile or run successfully.
Presentation (10%)	PPT Content (2.5%)	PPT is complete, well-structured, and visually clear. Includes all required items: student details, app overview, implemented features (including AI and additional features), and working video demo link (≤5 min). Content is concise and professional.	PPT includes all required items with minor missing details or minor design issues. Video link provided and relevant.	PPT is mostly complete but lacks clarity OR missing minor required elements (e.g. weak feature explanation or incomplete video documentation).	PPT is incomplete, poorly structured, or missing key sections (features or video unclear).	Student was absent for project presentation and demonstration without a valid reason. Clear evidence of plagiarism detected (disciplinable offence)
	Live Demonstration (2.5%)	Live demonstration is fully successful and smooth. All key features (authentication, CRUD, AI feature, additional features) are shown clearly and work without errors. Application flow is intuitive and matches proposal.	Live demonstration is mostly successful. Minor issues or delays, but all key features are shown and generally functional.	Live demonstration is partially successful. Some key features are shown but with noticeable errors, missing flows, or incomplete functionality.	Live demonstration is limited. Only basic features are shown; application has multiple errors or incomplete components.	
	Q&A (5%)	Answers all technical/code-related questions confidently, including reasoning behind implementation choices; explains key parts of the code accurately and in detail Present within the given time-limit.	Answers most technical/code-related questions confidently; gives reasonable explanation of code and logic. Present within the given time-limit.	Struggles with some code-related questions; explanations lack depth or clarity.	Unable to respond to most code or technical questions; shows limited understanding of own code.	